# Concept Note: FasalMitra

Smart crop and fertilizer advisor for small farmers · SDG 2 Zero Hunger

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## Why: the goal and the problem

I chose **SDG 2, Zero Hunger**, because food security starts on the farm. If farmers pick crops that suit their land and use fertilizer well, the same land can feed more people with less waste.

The problem I picked is that small farmers often choose crops and fertilizer by habit or by advice from neighbours and input sellers. A crop that does not suit the soil and weather lowers yield and income. Too much fertilizer wastes money and harms soil and water. Soil Health Cards list nutrient values (N, P, K and pH), but few farmers can turn those numbers into a decision.

The project also supports SDG 12 (less fertilizer wasted) and SDG 13 (less over-application, so less run-off and fewer emissions).

## What: the solution

FasalMitra is a web app. A user enters seven values: nitrogen, phosphorus, potassium, soil pH, temperature, humidity and rainfall. A machine-learning model suggests the three most suitable crops with a suitability score. For each crop the app shows which inputs sit inside that crop's normal range, and gives a plain fertilizer plan: what to add, what to skip, and roughly how much. It works in English and Hindi.

**How machine learning is used.** This is supervised classification. The model learns from labelled examples (soil and weather values paired with the crop that grows well there). I compared four algorithms (Decision Tree, KNN, Logistic Regression and Random Forest) on rows held back for testing, and kept the best one, which is a Random Forest [confirm the model name and accuracy after training on the final dataset]. The app is built with Python, scikit-learn and Flask.

**What makes it different.** Every recommendation is explained, so the farmer sees the reasons. There is no fertilizer-brand sponsorship, so the advice stays unbiased. It is bilingual from the start, and it can be retrained with a district's own data.

## Whom: who benefits

The primary users are **small and marginal farmers**. It also helps farmer producer organisations, Krishi Vigyan Kendras and extension workers, and NGO or CSR agriculture programmes. The early adopters are farmer groups and villages reachable through KVKs, NGOs and college agri clubs.

## Success and limits

Success would be tracked by the number of farmers advised, the fertilizer (kg per hectare) not applied after a “skip” recommendation, how often a local agriculture officer agrees with the plan, and whether farmers return the next season.

The limits are honest ones. The model is trained on a general benchmark dataset, not local farm records. The fertilizer quantities are estimates, and the app tells users to confirm them with their nearest KVK. It does not predict yield or price yet. The next steps are to add local data, predict yield with regression, and add WhatsApp and voice access in more languages.